A COMPOSITE OF LOCALIZED CHERN CLASSES EQUALS THE PUSHFORWARD OF THE RESTRICTED CLASSES ALONG THE INTERSECTION OF THE W_I

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FE7 (sources/stacks-chow/pinned-original-source.tex line 11026).

Allowed standard prerequisites (editorial).

- Localized Chern classes c'_p(Q) and the operators P'_p(Q) attached to a perfect object and a closed subset. Same chapter, Lemma labelled lemma-localized-chern-pre, invoked in the statement and cited at proof lines 11062-11067. Supplies the objects being composed and the case n = 1 identification.
- Base change behaviour of the localized class under restriction to a closed subscheme. Same chapter, Lemma labelled lemma-base-change-localized-chern-pre, cited at proof lines 11078-11082. Computes the restriction of c'_{p_n}(Q_n) to Z_{n-1} cap ... cap Z_1.
- The class C of the inclusion at infinity and its behaviour with pushforward. Same chapter, Lemma labelled lemma-gysin-at-infinity, cited at proof lines 11090-11091, and Lemma labelled lemma-homomorphism-pre, cited at proof lines 11108-11113. Provides the class C_{n-1} and the identity C_{n-1} o pushforward = id on Gysin images that reduces the lemma to a statement about images.
- Agreement of the restricted localized class with the bivariant Chern class of W_i -> W. Same chapter, Lemma labelled lemma-loc-chern-agree, cited at proof lines 11130-11134. Writes the restricted classes as restrictions of bivariant classes, so that commutation with i_infinity^* applies.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof.

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It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9K: sources/stacks-chow/pinned-original-source.tex, source lines 10204–10204. Same-chapter pointer into the author environment lemma at source line 10204; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0GUJ: sources/stacks-chow/pinned-original-source.tex, source lines 10678–10678. Same-chapter pointer into the author environment situation at source line 10678; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0GUI: sources/stacks-chow/pinned-original-source.tex, source lines 10276–10276. Same-chapter pointer into the author environment lemma at source line 10276; invoked at proof line 11076 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F9J: sources/stacks-chow/pinned-original-source.tex, source lines 10026–10026. Same-chapter pointer into the author environment lemma at source line 10026; invoked at proof line 11086 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0GUG: sources/stacks-chow/pinned-original-source.tex, source lines 9775–9775. Same-chapter pointer into the author environment lemma at source line 9775; invoked at proof line 11103 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FAR: sources/stacks-chow/pinned-original-source.tex, source lines 10132–10132. Same-chapter pointer into the author environment lemma at source line 10132; invoked at proof line 11117 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FB8: sources/stacks-chow/pinned-original-source.tex, source lines 10975–10975. Same-chapter pointer into the author environment lemma at source line 10975; invoked at proof line 11134 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $b : W \longrightarrow \mathbf{P}^1_X$ be a proper morphism of schemes. Let $n \geq 1$. For $i = 1, \ldots, n$ let $Z_i \subset X$ be a closed subscheme, let $Q_i \in D(\mathcal{O}_W)$ be a perfect object, let $p_i \geq 0$ be an integer, and let $T_i \subset W_\infty$, $i = 1, \ldots, n$ be closed. Denote $W_i = b^{-1}(\mathbf{P}^1_{Z_i})$. Assume*

(1) *for $i = 1, \ldots, n$ the assumption of Lemma 0F9K hold for b, Z_i, Q_i, T_i, p_i,*
(2) *$Q_i|_{W \setminus W_i}$ is zero, resp. isomorphic to a finite locally free module of rank $< p_i$ placed in cohomological degree 0,*
(3) *Q_i on W satisfies assumption (3) of Situation 0GUJ.*

Then $P'_{p_n}(Q_n) \circ \ldots \circ P'_{p_1}(Q_1)$ is equal to

$$(W_{n,\infty} \cap \ldots \cap W_{1,\infty} \to Z_n \cap \ldots \cap Z_1)_* \circ P'_{p_n}(Q_n|_{W_{n,\infty}}) \circ \ldots \circ P'_{p_1}(Q_1|_{W_{1,\infty}}) \circ C$$

in $A^{p_n + \ldots + p_1}(Z_n \cap \ldots \cap Z_1 \to X)$, resp. $c'_{p_n}(Q_n) \circ \ldots \circ c'_{p_1}(Q_1)$ is equal to

$$(W_{n,\infty} \cap \ldots \cap W_{1,\infty} \to Z_n \cap \ldots \cap Z_1)_* \circ c'_{p_n}(Q_n|_{W_{n,\infty}}) \circ \ldots \circ c'_{p_1}(Q_1|_{W_{1,\infty}}) \circ C$$

in $A^{p_n + \ldots + p_1}(Z_n \cap \ldots \cap Z_1 \to X)$.

Proof. Let us prove the statement on Chern classes by induction on n; the statement on $P_p(-)$ is proved in the exact same manner. The case $n = 1$ is the construction of $c'_{p_1}(Q_1)$ because $W_{1,\infty}$ is the inverse image of Z_1 in W_∞. For $n > 1$ we have by induction that $c'_{p_n}(Q_n) \circ \ldots \circ c'_{p_1}(Q_1)$ is equal to

$$c'_{p_n}(Q_n) \circ (W_{n-1,\infty} \cap \ldots \cap W_{1,\infty} \to Z_{n-1} \cap \ldots \cap Z_1)_* \circ c'_{p_{n-1}}(Q_{n-1}|_{W_{n-1,\infty}}) \circ \ldots \circ c'_{p_1}(Q_1|_{W_{1,\infty}}) \circ C$$

By Lemma 0GUI the restriction of $c'_{p_n}(Q_n)$ to $Z_{n-1} \cap \ldots \cap Z_1$ is computed by the closed subset $Z_n \cap \ldots \cap Z_1$, the morphism $b' : W_{n-1} \cap \ldots \cap W_1 \to \mathbf{P}^1_{Z_{n-1} \cap \ldots \cap Z_1}$ and the restriction of Q_n to $W_{n-1} \cap \ldots \cap W_1$. Observe that $(b')^{-1}(Z_n) = W_n \cap \ldots \cap W_1$ and that $(W_n \cap \ldots \cap W_1)_\infty = W_{n,\infty} \cap \ldots \cap W_{1,\infty}$. Denote $C_{n-1} \in A^0(W_{n-1,\infty} \cap \ldots \cap W_{1,\infty} \to Z_{n-1} \cap \ldots \cap Z_1)$ the class of Lemma 0F9J. We conclude the restriction of $c'_{p_n}(Q_n)$ to $Z_{n-1} \cap \ldots \cap Z_1$ is

$$\begin{aligned} &(W_{n,\infty} \cap \ldots \cap W_{1,\infty} \to Z_n \cap \ldots \cap Z_1)_* \circ c'_{p_n}(Q_n|_{(W_n \cap \ldots \cap W_1)_\infty}) \circ C_{n-1} \\ &= (W_{n,\infty} \cap \ldots \cap W_{1,\infty} \to Z_n \cap \ldots \cap Z_1)_* \circ c'_{p_n}(Q_n|_{W_{n,\infty}}) \circ C_{n-1} \end{aligned}$$

where the equality follows from Lemma 0GUG (we omit writing the restriction on the right). Hence the above becomes

$$\begin{aligned} &(W_{n,\infty} \cap \ldots \cap W_{1,\infty} \to Z_n \cap \ldots \cap Z_1)_* \circ c'_{p_n}(Q_n|_{W_n,\infty}) \circ \\ &\quad C_{n-1} \circ (W_{n-1,\infty} \cap \ldots \cap W_{1,\infty} \to Z_{n-1} \cap \ldots \cap Z_1)_* \\ &\quad \circ c'_{p_{n-1}}(Q_{n-1}|_{W_{n-1,\infty}}) \circ \ldots \circ c'_{p_1}(Q_1|_{W_{1,\infty}}) \circ C \end{aligned}$$

By Lemma 0FAR we know that the composition $C_{n-1} \circ (W_{n-1,\infty} \cap \ldots \cap W_{1,\infty} \to Z_{n-1} \cap \ldots \cap Z_1)_*$ is the identity on elements in the image of the Gysin map

$$(W_{n-1,\infty} \cap \ldots \cap W_{1,\infty} \to W_{n-1} \cap \ldots \cap W_1)^*$$

Thus it suffices to show that any element in the image of $c'_{p_{n-1}}(Q_{n-1}|_{W_{n-1,\infty}}) \circ \ldots \circ c'_{p_1}(Q_1|_{W_{1,\infty}}) \circ C$ is in the image of the Gysin map. We may write

$$c'_{p_i}(Q_i|_{W_{i,\infty}}) = \text{restriction of } c_{p_i}(W_i \to W, Q_i) \text{ to } W_{i,\infty}$$

by Lemma 0FB8 and assumptions (2) and (3) on Q_i in the statement of the lemma. Thus, if $\beta \in \mathrm{CH}_{k+1}(W)$ restricts to the flat pullback of α on $b^{-1}(\mathbf{A}^1_X)$, then

$$\begin{aligned} &c'_{p_{n-1}}(Q_{n-1}|_{W_{n-1,\infty}}) \cap \ldots \cap c'_{p_1}(Q_1|_{W_{1,\infty}}) \cap C \cap \alpha \\ &= c'_{p_{n-1}}(Q_{n-1}|_{W_{n-1,\infty}}) \cap \ldots \cap c'_{p_1}(Q_1|_{W_{1,\infty}}) \cap i_\infty^* \beta \\ &= c_{p_{n-1}}(W_{n-1} \to W, Q_{n-1}) \cap \ldots \cap c_{p_{n-1}}(W_1 \to W, Q_1) \cap i_\infty^* \beta \\ &= (W_{n-1,\infty} \cap \ldots \cap W_{1,\infty} \to W_{n-1} \cap \ldots \cap W_1)^* \left(c_{p_{n-1}}(W_{n-1} \to W, Q_{n-1}) \cap \ldots \cap c_{p_1}(W_1 \to W, Q_1) \cap \beta\right) \end{aligned}$$

as desired. Namely, for the last equality we use that $c_{p_i}(W_i \to W, Q_i)$ is a bivariant class and hence commutes with i_∞^* by definition. □